Physics Lecture 4: Quantum Physics

Introduction to the Quantum World

Quantum mechanics is the fundamental theory in physics that provides a description of the physical properties of nature at the scale of atoms and subatomic particles. It is the basis for understanding the behavior of matter and light at the microscopic level.

Key Concepts and Formulas

- **Quantization of Energy:** Energy is not continuous but exists in discrete packets called quanta. For a photon, the energy is proportional to its frequency.$E=hf$
  - E = energy of the photon
  - h = Planck's constant ($6.626\times10-34$ J·s)
  - f = frequency of the photon
- **Wave-Particle Duality:** Quantum objects exhibit properties of both waves and particles. The de Broglie wavelength for a particle is given by:$\lambda=ph=mvh$
  - λ = de Broglie wavelength
  - p = momentum
  - m = mass
  - v = velocity
- **Heisenberg's Uncertainty Principle:** It is impossible to simultaneously measure with perfect accuracy the position and momentum of a particle.$\Delta x\Delta p\geq2\hbar$
  - Δx = uncertainty in position
  - Δp = uncertainty in momentum
  - $\hbar$ = reduced Planck's constant ($2\pi h$)

Mathematical Example Problem: Photoelectric Effect

Problem: A metal surface with a work function of $\Phi=2.5$ eV is illuminated by light with a frequency of 1.0×1015 Hz. What is the maximum kinetic energy of the emitted photoelectrons?

Solution:

1. **Convert work function to Joules:** We need to use consistent units.$\Phi=2.5\text{ eV}\times(1.602\times10-19\text{ J/eV})=4.005\times10-19\text{ J}$
2. **Calculate the energy of the incident photons (**E_{photon}**):**$E_{photon}=hf=(6.626\times10-34\text{ J·s})\times(1.0\times1015\text{ Hz})E_{photon}=6.626\times10-19\text{ J}$
3. **Apply the photoelectric effect equation:** The maximum kinetic energy (K_{max}) is the difference between the photon energy and the work function.$K_{max}=E_{photon}-\Phi K_{max}=(6.626\times10-19\text{ J})-(4.005\times10-19\text{ J})K_{max}=2.621\times10-19\text{ J}$
   The maximum kinetic energy of the emitted photoelectrons is $2.621\times10-19$ J.